

Anand Chitikela

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PROFESSIONAL SUMMARY

Results-driven Senior Software Engineer and Lead AI Engineer with 12+ years of experience designing and building scalable backend systems, full-stack web applications, and AI-powered solutions. Expertise spans Python (FastAPI, Flask), PHP (Laravel, CodeIgniter), Node.js, React.js, and cloud-native architecture on AWS. Experienced in building production-grade Retrieval-Augmented Generation (RAG) systems that ground LLM responses in real data sources through semantic search and vector databases, as well as Agentic AI workflows using LangGraph — orchestrating autonomous, multi-step AI agents capable of reasoning, tool use, and adaptive decision-making across complex pipelines. Proven track record across telecom, healthcare, fintech, IoT, and CRM domains. Strong focus on maintainable code, security compliance (HIPAA, SOC2), and delivering high-impact solutions at scale.

TECHNICAL SKILLS

Languages:	Python, PHP, Node.js, JavaScript
Frontend:	React.js, React Native, Angular, HTML5, CSS3, Bootstrap, jQuery
Backend & APIs:	FastAPI, Flask, Express, LoopBack, Laravel, CodeIgniter, REST, SOAP, OpenAPI (Swagger)
AI / LLM:	LangChain, LangGraph, Agentic AI, RAG, OpenAI, Vector Databases
Databases:	PostgreSQL, MySQL, MS SQL Server
Cloud:	AWS (EC2, ECS, EKS, ECR, Lambda, API Gateway, S3, RDS, CloudWatch, CloudFormation, EventBridge), GCP, Google Calendar API
DevOps & CI/CD:	Docker, Kubernetes, GitHub Actions, Vercel
API Gateway:	Kong, Amazon API Gateway
CRM / Tools:	SugarCRM (Zend), Git, Bitbucket, GitLab, Jira, Confluence, VS Code, Cursor, PyCharm
Other:	RabbitMQ, Supervisor, Google Chrome Extensions, Pentaho ETL

PROFESSIONAL EXPERIENCE

Lead AI Engineer — CloudTern Solutions, India

Jul 2023 – Jun 2026

- Led the design and delivery of GenAI / RAG applications using LangChain and LangGraph.
- Architected end-to-end AI pipelines for insurance email ingestion (PolisIQ) and AI-powered search (Ansera).
- Deployed cloud-native microservices on AWS (Lambda, ECS, EKS) with Docker and Kubernetes.
- Ensured HIPAA and SOC2 compliance across all AI and backend services.
- Mentored junior engineers and served as a jury member for AI hackathons organised by Swecha Telangana.

Senior Software Engineer — JugruMS Technologies, India

Jul 2022 – Jun 2023

- Built and maintained scalable REST APIs with Python/FastAPI and integrated third-party services.
- Developed webhook-based integrations between Salesforce and Jira for workflow automation.
- Automated reporting and deployment pipelines using AWS services.

Software Engineer → Senior Software Engineer — KTree Computer Solutions, India

Mar 2014 – Jun 2022

- Delivered 10+ full-stack projects across telecom, fintech, IoT, tax, and CRM verticals.
- Developed multi-tenant MVNO portals, IoT monitoring dashboards, tax-filing platforms, and gift-card management systems.
- Customised and upgraded SugarCRM for 9 South African enterprises including JunkMail, Pnet, and Prudential.
- Integrated payment gateways, SOAP/REST APIs, RabbitMQ, and AWS S3 across multiple client portals.
- Built Google Chrome extensions for customer login and workflow automation.
- Won the 'Most Dependable Employee' award at KTree.

KEY PROJECTS

PolisiQ – Insurance & Claims Email Ingestion Tool

Technologies: Python, FastAPI, LangGraph, OpenAI, MySQL, React, GCP, Gmail API, Microsoft Graph API

- Architected a fully automated, end-to-end email ingestion pipeline using LangGraph's agentic AI framework, orchestrating five specialised agents — ingestion, classification, extraction, validation, and underwriting — each operating autonomously within a stateful, multi-step workflow.
- Integrated a LangGraph checkpointer so that any failed workflow run automatically resumes from its last successful checkpoint, ensuring resilience and eliminating duplicate processing.
- Connected FastAPI with both Gmail API and Microsoft Graph API to capture and process incoming insurance and claims emails in real time, regardless of the client's mail provider.
- Implemented an intelligent scoring engine that evaluates each submission and determines whether it is a viable fit for the insurance company, prioritising high-value leads and filtering out low-quality ones.
- Introduced a Human-in-the-Loop (HITL) review step, allowing operations staff to validate AI-extracted data before it is forwarded to underwriters or adjusters, maintaining accuracy and regulatory compliance.
- Developed a React portal for operations teams to manage submissions, track workflow status, review AI-generated insights, and approve or override underwriting decisions.

Ansera – AI-Powered Search Tool

Technologies: Python, FastAPI, LangChain, OpenAI, AWS Lambda, Docker, WordPress, Webflow, React, Google Calendar API

- Built a WordPress plugin that automatically ingests publicly available content from WordPress and Webflow pages and posts, indexes it using vector embeddings, and enables natural language Q&A; over the site's content.
- Developed a RAG-based search engine that processes PDFs and structured data, allowing admins to upload documents and videos that are made instantly searchable by the AI.
- Enabled admins to configure default questions, fine-tuning the AI's focus areas and improving the relevance of responses for end users.
- Delivered two distinct React CDN search interfaces — a full-width main search bar and a customisable floating search icon in the bottom-right corner — giving site owners flexible embedding options.
- Extended the platform with MCP (Model Context Protocol) capabilities, allowing users to schedule a meeting directly with the sales team through the AI, powered by Google Calendar API integration.
- Deployed the entire backend as a serverless, cloud-native architecture on AWS Lambda to minimise infrastructure overhead and optimise cost at scale.

Skywire Networks – Salesforce & Jira Integration

Technologies: Python, Flask, AWS, Jira REST API, Salesforce API, PlantUML, Excel

- Integrated Salesforce APIs to automate reporting and streamline business workflows across sales and operations teams.
- Documented complex business processes as PlantUML diagrams, providing clear visual references that guided both development and stakeholder sign-off.
- Consumed Jira REST APIs to fetch active sprint data by board ID and generated structured Excel reports summarising ticket status for engineering managers and stakeholders.
- Automated sprint-closing report generation and email distribution using scheduled AWS jobs, reducing manual reporting effort by approximately 80%.

Syntegra – MVNE Platform

Technologies: Python, Flask, PHP, MySQL, PostgreSQL, Docker, AT&T; APIs, Zoho Ticketing, Swagger (OpenAPI)

- Built policy and usage engines that enforce plan rules, throttle data speeds at defined thresholds, and automatically block data once usage limits are reached.
- Integrated AT&T; carrier APIs, Zoho Ticketing, and multiple third-party services within a microservices architecture to support end-to-end MVNE operations.
- Developed a multi-tenant portal enabling MVNOs to create new subscriptions and port-ins for a major North American carrier, with real-time carrier connectivity via a Node.js middleware layer.
- Built analytics dashboards displaying active and suspended subscription counts broken down by plan and MVNO, along with detailed admin reports.
- Implemented multi-factor authentication using Google Authenticator to secure portal access.

- Standardised all internal and external API contracts with Swagger (OpenAPI) documentation, improving developer onboarding and cross-team consistency.

CloudTruss – MVNE Admin Portal

Technologies: Angular, TypeScript, MySQL, Docker, Kubernetes

- Developed comprehensive admin and customer-facing portals supporting multiple MVNO tenants, each with isolated role-based access controls.
- Managed the full customer lifecycle — from onboarding and subscription management to suspension and termination — within a scalable microservices architecture.

Vivio – Healthcare Bridge Fills & Adjudication Platform

Technologies: Python, FastAPI, SuiteCRM, RxSense API

- Designed and implemented a bridge fills architecture that ensures uninterrupted medication supply to patients by bridging the gap between the healthcare provider and the insurance company while a Vivio prescription is being processed and dispensed.
- Built the RxSense adjudication integration to automate claims processing, accurately calculating reimbursements and submitting approved payments to healthcare providers.

ACN Fixed Line – Admin & Customer Portal

Technologies: PHP, React.js, CodeIgniter, AWS S3, RabbitMQ, Supervisor

- Replaced a legacy Delphi-based billing and invoice system with a modern, scalable web platform serving customer service agents and end customers across 10 European countries.
- Delivered a full suite of operational modules: payment gateway integration with mass direct debit and offline payment support; a provisioning module for connection termination and resumption; a dunning engine that stages overdue accounts through escalating actions up to termination; a ticketing system for complaint assignment and tracking; and an automated invoicing pipeline that bundles offline invoice data into a single print-vendor file for postal delivery.

Bhaswa – Tax Filing Platform

Technologies: PHP, Laravel 5.5, jQuery, Ajax, AWS S3

- Built dual customer and admin portals for a tax processing service, enabling clients to register, upload information sheets and documents, and track submission status.
- Implemented TPA (Tax Processing Associate) assignment logic, automatically routing each customer to the appropriate associate based on the selected service.
- Used AWS S3 as the communication bridge between the customer and admin portals, keeping a unified database to eliminate data duplication.
- Developed a Google Chrome extension to streamline customer login management directly from within the application.
- Integrated a payment gateway to handle service fee collection at the point of submission.

Spendmex – Gift Card Management Platform

Technologies: PHP, Laravel, jQuery, Ajax, SOAP/REST APIs, AWS S3

- Automated card operations — activation, recharge, PIN change, status update, and balance inquiry — by integrating with a bank's WSDL via a SOAP client.
- Built three-tier role-based portals: an administrator portal for card and distributor management; a corporate portal shared by distributors and master clients for card assignment and request submission; and a customer portal for new card requests.
- Implemented a transaction management system that logs every card operation at both the client and user level, providing a full audit trail for all requests processed through the corporate portal.

Regus – IoT Remote Weather Monitoring

Technologies: React.js, Node.js, Express.js

- Built a real-time remote weather monitoring portal that aggregates environmental data streamed from IoT devices deployed across multiple locations.
- Created device-specific modules and live temperature trend graphs, giving operations teams instant visibility into conditions at each remote site.

WebActe – Mobile Document & Notification App

Technologies: React Native

- Developed the frontend of a React Native mobile application covering file management, push notifications, and customer comment threads, integrating cleanly with a backend maintained by a separate team.

SugarCRM – CRM Customisation & Upgrades

Technologies: PHP, jQuery, Ajax, SugarCRM (Zend), Pentaho ETL

- Customised, configured, and upgraded SugarCRM deployments for nine South African enterprises — JunkMail, Bosun, IDA, Pnet, Sentech, Vision Medicals, Rantech, Prudential, and Servest.
- Executed full version migrations from SugarCRM 6.x to 7.x, preserving all existing customisations and migrating data using Pentaho ETL.
- Delivered bespoke features including new modules, dashboards, reports, web-to-lead capture forms, automated birthday alerts, and campaign management.
- Integrated each CRM with external .NET and Java applications via REST APIs to enable seamless cross-platform data exchange.

EDUCATION

Bachelor of Technology – Computer Science and Engineering

Joginpally B.R. Engineering College (JNTU Hyderabad), 2014

ACHIEVEMENTS & INTERESTS

- Won the Participation Award from Chief Minister of Andhra Pradesh N. Kiran Kumar Reddy for volunteering at COP-11 (2012).
- Won the 'Most Dependable Employee' Award at KTree Computer Solutions.
- Serving as a jury member for AI-focused hackathons and technical events organised by Swecha Telangana; actively mentoring students on their projects.
- Interests: Reading, Chess.